Book: Fundamentals of Mechatronics, by M. Jouanh

B.3 Frequency Response

The **frequency response** of a system is the response of the system to a periodic input signal (such as a sinusoidal input as a function of the frequency of the sinusoid). The frequency response is normally shown as two plots: a magnitude plot and a phase plot. In the **magnitude plot**, the ratio of the output of the input is plotted as a function of frequency. Normally, the magnitude plot is displayed in units of **dB** where 1 dB = 20 log (output to the input). In the **phase plot**, the angle between the output and the input signals is displayed as a function of frequency. To illustrate the process of generating the frequency response plots, let us consider the transfer function of a low-pass filter, which is given by $G(s) = 1/(1 + \tau s)$ (see Section 7.10). We replace the variable s in the transfer function with $j\omega$. Performing this, we obtain

$$G(j\omega) = \frac{1}{1 + j\omega\tau} \tag{B.8}$$

To get rid of the complex quantity in the denominator, we multiply the numerator and the denominator of Equation (B.8) by the complex conjugate of $1 + j\omega\tau$, which is $1 - j\omega\tau$. This gives

$$G(j\omega) = \left(\frac{1}{1 + j\omega\tau}\right)\frac{1 - j\omega\tau}{1 - j\omega\tau} = \frac{1 - j\omega\tau}{1 + \omega^2\tau^2} = \frac{1}{1 + \omega^2\tau^2} - \frac{j\omega\tau}{1 + \omega^2\tau^2} \tag{B.9}$$

The **magnitude** of the transfer function is then obtained by taking the square root of the sum of the squares of the real and imaginary parts in Equation (B.9). This gives

$$|M(j\omega)| = \sqrt{\left(\frac{1}{1 + \omega^2\tau^2}\right)^2 + \left(\frac{\omega\tau}{1 + \omega^2\tau^2}\right)^2} = \frac{1}{\sqrt{1 + \omega^2\tau^2}} \tag{B.10}$$

The **phase angle** is obtained from evaluating the inverse tangent of the ratio of the imaginary part to the real part or

$$\varphi = \tan^{-1}\left(\frac{-\tau\omega}{1}\right) = -\tan^{-1}(\tau\omega) \tag{B.11}$$

Equations (B.10) and (B.11) can be plotted as functions of the angular frequency ω to obtain the frequency response plot of the transfer function. These plots are given in Figures B.7 and B.8, respectively, where the magnitude plot is displayed in dB units.

From Equation (B.10), we see that when $\omega\tau$ is very small; the magnitude is approximately equal to 1. Since the logarithm of 1 is zero, then the magnitude in dB units is equal to 0 for small values of $\omega\tau$. Similarly, when $\omega\tau$ is very large, Equation (B.10) can be approximated by $(\omega\tau)^{-1}$. Hence, the magnitude in dB is

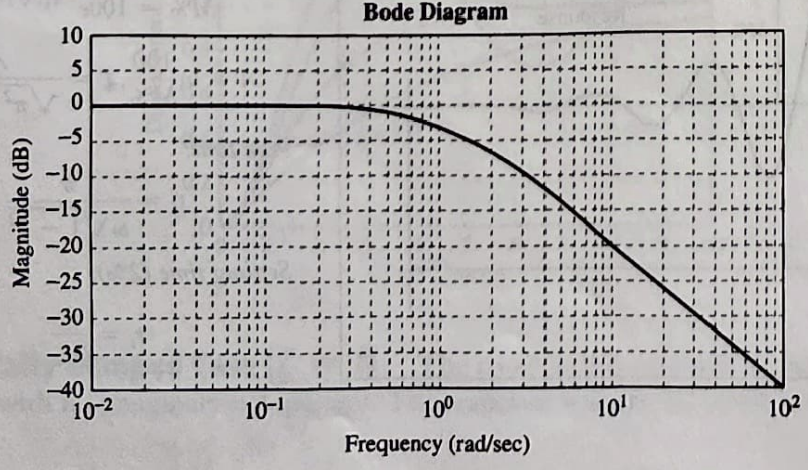


Figure B.7 Magnitude plot of a first-order system

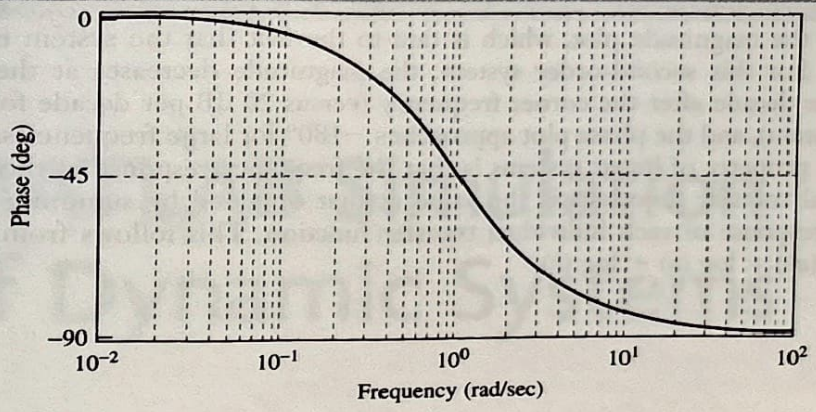


Figure B.8

Phase plot of a first-order system

equal to $-20 \log \omega - 20 \log \tau$. Thus, the slope of the frequency magnitude plot is -20 dB per decade, where a decade is a 10 times increase or decrease in the frequency. When $\omega = 1/\tau$, or is at the **corner frequency**, the magnitude has a value of -3 dB. Similarly for the phase plot, we can see from Equation (B.11) that when $\omega\tau$ is very small, its inverse tangent is close to 0°, while when $\omega\tau$ is very large, its inverse tangent is close to $-90°$. A **negative phase angle** means that the output signal lags the input signal, while a **positive phase angle** means that the output signal leads the input signal. The relationship between the phase angle (φ in degrees) and the **lead/lag time** (Δt in seconds) at any particular frequency (ω in rad/s) is given by

$$\Delta t = \frac{\varphi\pi}{180\omega} \tag{B.12}$$

As an example, at a frequency of 10 rad/s, the output signal from the low-pass filter will lag the input signal by 0.147 seconds.

MATLAB has two commands for generating the frequency response plots. They are the ***Bode*** and ***Bodemag***. The *Bode* command generates both the magnitude and phase plots, while the *Bodemag* generates the magnitude plot. Figure B.9 shows

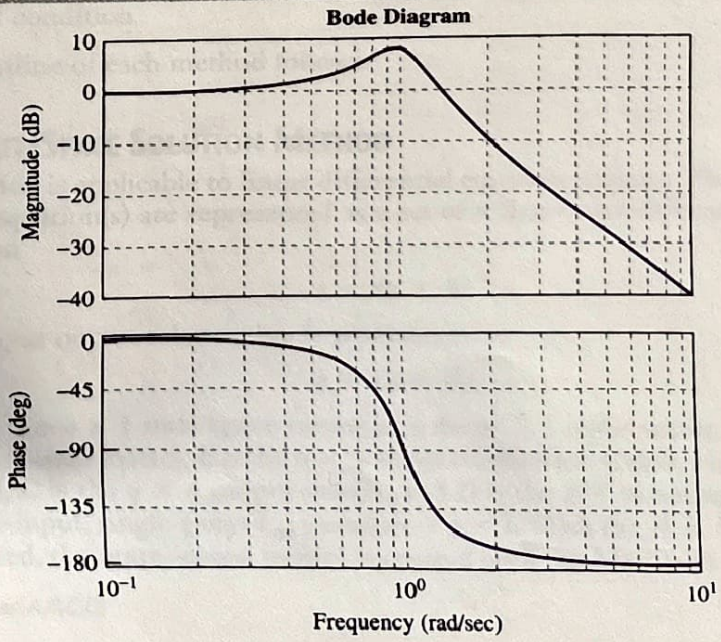


Figure B.9

Frequency response plot for an underdamped second-order system

the Bode plot for the second-order system given by Equation (B.3). Notice the hump in the magnitude plot, which is due to the fact that the system is underdamped. For this second-order system, the magnitude decreases at the rate of 40 dB per decade after the corner frequency (versus 20 dB per decade for a first-order system), and the phase plot approaches $-180°$ for large frequencies.

One property of linear systems is that the frequency response of the product of several transfer functions is the same at that obtained by summing the frequency response of each individual transfer function. This follows from the fact that $\log(ab) = \log(a) + \log(b)$.